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Radiotherapy and Oncology 36 (1995) 94–100

RADIOTHERAPY & ONCOLOGY

# Radionecrosis of the mandibula: a retrospective analysis of the incidence and risk factors

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Received 20 June 1994; revision received 10 May 1995; accepted 6 June 1995

**Abstract**

We analysed the incidence of a mandibular osteonecrosis (ON) in 189 patients irradiated for a cancer of the oral cavity, or of the oropharynx or epipharynx, with a total target dose of at least 60 Gy between 1980 and January 1994. Target doses per fraction were between 2.0 and 1.2 Gy, number of sessions per week was between 5 and 10, and total target doses were between 60 and 78.2 Gy. No instance of ON has been observed after target doses between 60 and 65 Gy. Cumulative incidence of an ON needing a mandibular resection was: 24.8% in patients treated with a mandibular dose per fraction between 2.00 and 2.22 Gy and a total mandibular dose between 66.00 and 79.70 Gy; 19.6% in patients treated with a mandibular dose per fraction between 1.80 and 1.90 Gy and a total mandibular dose between 69.00 and 75.60 Gy, 2.2% in patients treated with two fractions of 1.20 Gy per day for 5 days in the week with or without simultaneous application of *cis*-platinum and a total mandibular dose between 75.20 and 82.00 Gy. Dose per fraction, association of the tumour with bone, and the volume of the horizontal ramus of the mandibula irradiated with a high dose were observed to be significant risk factors.

*Keywords:* Radioosteonecrosis; Mandibula; Hyperfractionation; Tolerance dose

## 1. Introduction

Radionecrosis of the mandibula can be a serious complication after radiotherapy (rt) in patients with cancers of the oral cavity or of the oro- or nasopharynx. Published figures of the incidence of mandibular radionecrosis vary widely from 4 to more than 50% with mandibular resection necessary in approximately half of these patients [1,5,7,8,12] (all from MDA Hospital at Houston), [4,9], reviewed in [11]. Most of these reports give only crude incidence rates and no sufficient details on the techniques of rt. In an analysis of patients, who were treated between 1964 and 1975 at MDA hospital and survived at least 5 years without recurrence, Larson et al. 1982 [5] observed in 56.3% a grade 4 osteonecrosis. According to several published dose-response curves [10], and some figures and statements in the 2nd and 3rd editions of the Textbook of Radiotherapy by Fletcher [3], an important percentage of the patients reported in the papers of Murray et al. [7,8] and of Larson et al. [5] have received mandibular doses per fraction significantly higher than 2 Gy and also total doses well above 70 Gy. Some of these patients had interstitial and/or intraoral cone therapy in addition to the external rt.

Because there are only relatively few data on the risk of mandibular necrosis after rt with modern techniques, we analyzed, retrospectively, the records of our patients treated between 1980 and 1993.

## 2. Patients and methods

### 2.1. Patients

Since the early 1970s, all patients with cancer of the head and neck have been examined in a joint clinic by the head neck surgeon together with a radiotherapist both before treatment and during the regular follow-up

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*SSDI* 0167-8140(95)01583-3

after treatment. All our patients with bone exposure lasting more than approximately 3 months or other signs of radionecrosis are referred to the dental department of the clinic for Oral and Maxillofacial Surgery. A preliminary analysis of the records showed that osteonecrosis occurred only in patients with malignant tumors of the oral cavity or of the oro- or nasopharynx after a target dose exceeding 60 Gy applied at 2.0 or 1.8 Gy per fraction 5 times per week. Computerized tomography (CT) for irradiation planning was installed in 1979. Therefore, we reviewed the case histories of all 189 patients who were treated for a malignant tumour of the oral cavity, the oropharynx or the nasopharynx, with at least 60 Gy alone or combined with surgery (s) and/or chemotherapy (ch) between January 1, 1980 and January 1994, and who had a complete remission of the primary tumour. In 52 patients, the target dose was between 60 and less than 66 Gy. In these patients, no instance of osteonecrosis occurred. The detailed study was therefore limited to the 137 patients with a target dose of at least 66 Gy. A summary of the patient characteristics is given in Table 1. Further details in patients with osteonecrosis are given in Tables 2 and 3 in the results section. Only one patient has been lost from follow-up 8 years after treatment with no signs of recurrence or osteonecrosis.

A tumor has been classified as closely associated with the mandibula if at least one of the following conditions were met:

(1) localization of the tumour in the lateral part of the floor of the mouth or in the lower gingiva or in the retromolar trigone;

(2) tumour extension abutting on or infiltrating the bone as shown in the CT. All patients had a CT scan. In the group treated with 2.0 Gy per fraction, there was one patient with primary rt and signs of bone infiltration and two patients with postoperative irradiation (including a partial or a marginal mandibular resection)

Table 1
Characteristics of patients irradiated with a target dose of ≥66 Gy for carcinoma of the oral cavity or of the naso- or oropharynx and analyzed for radioostonecrosis (USZ: treatment period 1/80–1/94, analysis 1/95)

| | 1980–1987 | since 1988 | since 1988 |
|---|---|---|---|
| fractionation | 2.0 Gy/fraction<br>1 fraction/day<br>5 fractions/week | 1.2 Gy/fraction<br>2 fraction/day<br>10 fractions/week | 1.8 Gy/fraction<br>1-2 fraction/day<br>5–7 fractions/week |
| mean total target dose (Gy) | 69.4 | 74.5 | 68.9 |
| (range) | (66.0–70) | (72.0–78.4) | (66.4–72.4) |
| floor of mouth | 7 (11%) | 4 (9%) | 1 (4%) |
| lower gingiva | 2 (3%) | — — | 1 (4%) |
| mobile tongue | 1 (2%) | — — | 1 (4%) |
| base of tongue | 9 (15%) | 11 (23%) | 8 (28%) |
| lateral oropharynx | 22 (36%) | 19 (40%) | 15 (53%) |
| retromolar tongue | 3 (5%) | 1 (2%) | — — |
| nasopharynx | 17 (28%) | 12 (26%) | 2 (7%) |
| T1 + 2 | 30 (49%) | 12 (26%) | 20 (71%) |
| T3 | 18 (30%) | 16 (34%) | 7 (25%) |
| T4 | 13 (21%) | 19 (40%) | 1 (4%) |
| N0 | 27 (44%) | 13 (28%) | 17 (39%) |
| N+ | 34 (56%) | 34 (72%) | 11 (79%) |
| association with bone | | | |
| yes | 20 (33%) | 21 (45%) | 6 (21%) |
| no | 38 (62%) | 26 (55%) | 22 (79%) |
| infiltration of bone | 3 (5%) | — — | — — |
| teeth in exposed mandibula | 16 (34%) | 6 (13%) | 7 (25%) |
| edentulous | 24 (51%) | 25 (52%) | 11 (39%) |
| teeth outside high dose region | 7 (15%) | 16 (35%) | 10 (36%) |
| unilateral irradiation | 22 (36%) | 3 (6%) | 9 (32%) |
| bilateral irradiation | 39 (44%) | 44 (94%) | 10 (68%) |
| follow-up (years) mean | 5.1 | 3.1 | 2.8 |
| $n \geq 3$ years | 23 | 20 | 7 |
| fraction of mandibula included | | | |
| score ≤ 2 | 12 (20%) | 3 (6%) | 1 (4%) |
| score 3–4 | 7 (12%) | 9 (19 %) | 8 (28%) |
| score ≥ 5 | 42 (68%) | 35 (75%) | 19 (68%) |
| surgery and postoperative rt | 3 (5%) | 2 (4%) | 21 (75%) |
| rt alone | 58 (95%) | 45 (96%) | 7 (25%) |

Table 2
Cumulative actuarial incidence of osteonecrosis (ON) of the mandible after external rt for cancer of the oral cavity or the oro- or nasopharynx with a target dose of ≥66 Gy (USZ: 1/1980–1/1994; analysis 1/1995) ($n$ = number of patients)

| Patient group | | Years after treatment | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 | 8 |
| **total group ($n$ = 137)** | | | | | | | | |
| $TD_{max}$: 75.1 (mean)/ 66.0–82.3 | $n$ in follow up | 131 | 103 | 79 | 55 | 35 | 28 | 12 |
| (range) | total $n$ of ON | 8 | 12 | 14 | 16 | 18 | 19 | 20 |
| $TD_{min}$: 70.5/ 60.0–78.4 | % actuarial incidence of ON | 5.9 | 9.1 | 11.3 | 14.2 | 18.6 | 21.5 | 28.0 |
| | total $n$ of ON of grade ≥3 | 5 | 8 | 10 | 12 | 13 | 13 | 14 |
| | % actuarial incidence of ON grade ≥ 3 | 6.1 | 8.3 | 11.2 | 13.4 | 13.4 | 20.6 | |
| (2 × 1.2 Gy/day for 5 days/week or 1 × 1.8 Gy/day or 2 × 1.8 Gy or 1 × 1.8 Gy + 1 × 1.5 Gy/day for 5–7 fractions/week or 2.0 Gy/day for 5 days/week or 3 × 1.6 Gy/day with 72 Gy/ 5 weeks) | | | | | | | | |
| **2 × 1.2 Gy/day, 5 days/week** | | | | | | | | |
| $TD_{max}$: 77.8/74.4–82.3 | $n$ in follow-up | 46 | 35 | 24 | 18 | 8 | 4 | |
| $TD_{min}$: 74.5/72.0–78.4 | total $n$ of ON | 4 | 5 | 5 | 6 | 6 | 7 | |
| $D_{max}$ 1.23/1.20–1.26 | % actuarial incidence of ON | 8.5 | 11.1 | 11.1 | 21.0 | 21.0 | 40.7 | |
| $D_{min}$ 1.2/1.2–1.2 | total $n$ of ON of grade ≥3 | 1 | 1 | 1 | 1 | 1 | 1 | |
| | % actuarial incidence of ON grade ≥3 | 2.2 | 2.2 | 2.2 | 2.2 | 2.2 | 2.2 | |
| **2.0 Gy/day, 5 days/week** | | | | | | | | |
| $TD_{max}$: 74.5/ 66.0–79.7 | $n$ in follow-up | 57 | 46 | 39 | 33 | 27 | 24 | 12 |
| $TD_{min}$: 68.0/60–72.0 | total $n$ of ON (all of grade ≥3) | 3 | 6 | 7 | 8 | 9 | 9 | 10 |
| $D_{max}$: 2.18/2.0–2.2 | % actuarial incidence of ON | 4.9 | 10.3 | 12.4 | 15.0 | 18.0 | 18.0 | 24.8 |
| $D_{min}$: 2.0/ 1.8–2.0 | | | | | | | | |
| **1.8 Gy/day, 5–7 fractions/week** | | | | | | | | |
| $TD_{max}$: 72.2/66.6–75.6 | $n$ in follow-up | 28 | 21 | 16 | 4 | | | |
| $TD_{min}$: 69.0/66.0–72.0 | total $n$ of ON (all of grade ≥3) | 0 | 0 | 1 | 2 | | | |
| $D_{max}$: 1.88/1.8–1.9 | % actuarial incidence of ON | 0 | 0 | 6.2 | 19.6 | | | |
| $D_{min}$: 1.8/1.8–1.8 | | | | | | | | |

$D_{max}/D_{min}$: Maximal/minimal dose per fraction to the mandible (Gy)
$TD_{max}/TD_{min}$: Maximal/minimal total dose to the mandible (Gy)
The total group includes one patient with an ON after treatment with 3 × 1.6 Gy per day and 72 Gy total target dose in 5 weeks.

with histological signs of mandibular infiltration. In the other patients, no evidence of mandibular involvement was found. In patients with signs of bone involvement, primary surgery and postoperative rt with a somewhat lower dose is preferred.

### 2.2. Dental care before rt

Prior to rt, all patients are seen in the dental clinic of the Department for Oral and Maxillofacial Surgery. Edentulous patients have a radiographic examination and, if necessary, surgical removal of infected retained root tips or alveolar hyperplasia. The rest of the patients have dental treatment following the guidelines of Daly et al. [2]. Patients are instructed about dental care and are supplied with an individual fluoride carrier. During the period of rt, these patients have one or two further visits to the dental hygienist.

In 1974/75, the pretherapeutic dental evaluation and treatment was changed from a more radical extraction of all or nearly all of the remaining teeth to a more conservative treatment according to the guidelines of Daly et al. [2]. This change was induced by the publication of the detailed guidelines in the second edition of Textbook of Radiotherapy by G. Fletcher in 1973 and these guidelines have since been adhered to.

### 2.3. Definition and treatment of osteonecrosis

Osteonecrosis (ON) was defined and graded in the following manner: ON grade 1, bone exposure without signs of infection and persisting for at least 3 months; ON grade 2, bone exposure with signs of infection or sequester and without the signs of grade 3–5; ON grade 3, bone necrosis treated with mandibular resection with a satisfactory result; ON grade 4, bone necrosis with

Table 3
Incidence of osteonecrosis (ON) of the mandible after external rt for cancer of the oral cavity, the oro- or nasopharynx (USZ: 1/1980–1/1994; analysis 1/1995)

| Patient group | Number (*n*) treated[a] | *n* with ON | *n* with ON ≥ grade 3 |
|---|---|---|---|
| **2 × 1.20 Gy target dose/day, 6 h interval, 5 days/week, total dose 72.00–78.40 Gy; maximal dose to the mandible: 74.4–82.3 Gy** | 47 | 7 | 1 (fatal) |
| edentulous | 25 | 5 | [b] |
| no teeth in exposed bone | 16 | 1 | |
| teeth in exposed bone | 6 | 0 | 0 |
| mandibular volume exposed: | | | |
| score ≥5 | 35 | 6 | 1 |
| score ≥4 | 12 | 1 | 0 |
| close association with bone | | | |
| yes | 21 | 5 | 1 |
| no | 26 | 2 | 0 |
| **1.80 Gy target dose/day, 5 or 6 days/week, total dose 66.40–72.00 Gy; maximal dose to the mandible: 66.6–75.60 Gy** | 28 | 2 | 2 |
| edentulous | 11 | 1 | 1 |
| no teeth in exposed bone | 10 | 0 | 0 |
| teeth in exposed bone | 7 | 1 | 1 |
| mandibular volume exposed | | | |
| score ≥5 | 19 | 2 | 2 |
| score ≤4 | 9 | 0 | 0 |
| close association with bone | | | |
| yes | 6 | 1 | 1 |
| no | 22 | 1 | 1 |
| **2 Gy target dose/day, 5 days/week, total target dose 66.00–72.00 Gy; maximal dose to the mandible: 66.00–79.7 Gy** | 61 | 10 | 10 |
| edentulous* | 24 | 3 | 3 |
| no teeth in exposed bone | 7 | 0 | 0 |
| teeth in exposed bone | 16 | 6 | 6 |
| mandibular volume exposed** | | | |
| score ≥5 | 42 | 10 | 10 |
| score ≤4 | 19 | 0 | 0 |
| close association with bone*** | | | |
| yes | 23 | 7 | 7 |
| no | 38 | 3 | 3 |

[a]One patient with ON after 3 × 1.6 Gy/day omitted.
[b]Dental extractions before rt; ON outside of region with dental extraction.
*$P = 0.057$; **$P = 0.02$; ***$P = 0.051$. All the other differences are not significant.

persisting problems despite mandibular resection; ON grade 5, death due to ON.

After clinical diagnosis of a bone necrosis, radiographic examination and conservative treatment was applied initially: multiple daily saline irrigations, antibiotics during infectious episodes, topically applied antiseptics, gentle sequestrectomy and removal of visibly loosened bone elements as well as treatment with hyperbaric oxygenation.

Mandibular resection and reconstructive interventions were performed in patients in whom the bone necrosis proceeded with intractable pain and/or pathological fracture.

### 2.4. Analysis of the records; radiotherapy methods

We analyzed the localization and TNM stage of the tumor, age and sex of the patient, dental extractions before and after rt, treatment modality and details of rt, as well as the incidence and details of ON. There were too few notes for an evaluation of the compliance of the patient with the dental care program and a quantifica-

tion of the alcohol intake. But in the majority of patients a significant use of alcohol was noted.

### 2.5. Radiotherapy

In 1979, the first whole body CT-scanner for treatment planning was installed in the Department of Radiation Oncology.

All patients had photon rt from a 6 MV linear accelerator. Nine patients received an initial series of 36 Gy ($\gamma$ ray) of 60-cobalt and five patients received 10 Gy with 12–16 MeV electrons as a boost. All patients had CT-based treatment plans calculated with a GE RT-plan, except during 1980 and 1981, when isodose calculations were performed by means of the Sterling program on a central computer. Target dose was prescribed as an isodose, including the target volume as marked on CT slices. Up to 1987, target dose per fraction was 2.00 Gy prescribed mostly on a 90% isodose, 5 times per week. Since 1988 routine target dose per fraction has been 1.80 Gy prescribed as a 95% isodose, 5 or 6 times per week (usually with a second session on Friday after an interval of 6 h). Twelve patients had an accelerated rt with a concomitant boost technique: 1.80 Gy 5 times per week on the large volume and two sessions with 1.50 Gy on the boost volume after an interval of 6 h. Since 1992, we modified our concomitant boost technique, applying two fractions per day, 5 days per week, in the second half of the rt. Since 1988, patients with advanced tumours and primary rt were treated with a hyerfractionated rt and simultaneous application of cisplatinum: two fractions of 1.20 Gy with an interval of at least 6 h, 5 days per week, and a total dose of between 72.0 and 78.2 Gy, and 20 mg/m$^2$ cisplatinum as rapid intravenous infusion, approximately 30 min before the second session on the 5 days of the first and the fifth weeks of rt. One patient has been irradiated with 3 × 1.6 Gy per day on 5 days per week, with a 4-h interval and a total dose of 72 Gy in 5 weeks. This patient developed an ON which necessitated mandibular resection. Head fixation with a mask, simulation and at least weekly check films were used in all patients.

All treatment protocols, isodose plans and simulation as well as check films were analyzed. Volume of mandibula irradiated with the target dose, maximal and minimal total dose and dose per session have been checked.

### 2.6. Classification of the mandibular volume included in the high dose volume

Analyzing the data, the following classification of the fraction of the mandibula included in the volume of the target dose was used: 1, ramus ascendens; 2, part of the ramus ascendens plus angle; 3, whole ramus ascendens plus angle; 4, part of ramus ascendens plus less than half of ramus horizontalis; 5, whole ramus ascendens plus less than half of ramus horizontalis; 6, more than half of but not the whole ramus horizontalis plus angle and part of ramus ascendens; 7, whole ramus horizontalis (including chin region) plus angle. Furthermore, uni- and bilateral irradiation has been distinguished.

### 2.7. Data analysis

Storage and analysis of data has been performed with the Medlog data management system (version 91.8). Intragroup comparison of proportions were done by the chi-square test. Comparison of incidence curves has been done with the Mantel-Haenszel log rank test. The actuarial cumulative incidence of ON has been calculated according to the Kaplan-Meier method. The proportional hazards regressions model (Cox model) was used to estimate the relative risk of ON with adjustment for confounding variables such as dental extractions, relation of tumour to bone, T stage, total dose and dose per fraction, length of follow-up, proportion of the mandibula included in the target dose and chemotherapy. Because at least 10 events are necessary for one variable, this type of analysis can be considered only with some reservation.

There was only one patient treated with 3 × 1.6 Gy per day: this patient is included only in the group of all patients studied but omitted in the analysis of the subgroups.

## 3. Results

### 3.1. Characteristics of patient groups

Table 1 shows some important characteristics of the different patient groups: the localization of the primary tumours does not show significant differences between the groups. Patients treated with 2 × 1.2 Gy per day had a significantly higher fraction of T3 and T4 primaries as well as a higher fraction of involved lymph nodes compared with the other groups. In the group treated with 2.0 Gy per fraction, there were some more patients with teeth in the exposed part of the mandibula ($P = 0.03$). The fraction of the mandibula included in the high dose volume was significantly higher in the group treated with 2 × 1.2 Gy per day. Length of follow-up was significantly higher in the group treated with 2.0 Gy per fraction. Sixty-two percent of the patients treated with 2 × 1.2 Gy per day had simultaneous chemotherapy with CDDP; 5% of the patients treated with 2 Gy per fraction had induction chemotherapy.

### 3.2. Incidence of radioosteonecrosis (ON)

No instance of ON has been observed in 52 patients

with total target doses between 60.00 and 65.99 Gy applied with 2 Gy per fraction, 5 times per week, or with 1.8 Gy per fraction, 5–7 times per week, or a concomitant boost technique with 1.8 and 1.5 Gy. Maximal mandibular doses in these patients were between 64.00 and 72.10 Gy, applied at a maximal dose per fraction of between 2.22 and 1.89 Gy. All calculations concerning the relative frequency of ON are, therefore, limited to the 137 patients with total target doses of 66.00 Gy or higher. Total mandibular doses in these patients were between 66.0 and 82.3 Gy. The crude rate of ON was 14.6% (20/137). The crude rate of the various grades of ON was: 1/137 (0.7%) grade 1; 5/137 (3.6%) grade 2; 6/137 (4.4%) grade 3; 4/137 (3.9%) grade 4; and 4/137 (2.9%) grade 5. Two of the four patients with fatal ON died due to inanition and infectious complications without surgical intervention, and the other two died of cardiorespiratory arrest after repeated surgery for ON. Heavy abuse of alcohol and personal neglect was an important contributory factor to the fatal outcome in these patients. Two of these patients had a biopsy several weeks before death and the other two had a postmortem examination confirming an osteonecrosis and the absence of tumour.

Forty percent of the ON were diagnosed in the first year after rt, 60% in the first 2 years and 20% were diagnosed between 5 and 8 years after rt. After a follow-up of 8 years, the actuarial incidence of ON was 28% for all grades and 20.6% for grade 3 or higher.

### *3.3. Influence of dose, fractionation, dental extractions, volume of mandibula exposed to high dose and relation of tumour to bone (Tables 2 and 3)*

An univariate analysis showed the following significant risk factors for ON of at least grade 3: dose per fraction ($P = 0.035$ comparing 2 Gy with 2 × 1.2 Gy), close association of the tumour to the bone ($P = 0.051$), fraction of the mandibula included in the high dose volume ($P = 0.026$) comparing the score $<4$ with 4 or higher. T stage was not a significant factor. In a multivariate analysis, however, only dose per fraction and a close association of the tumour with the bone remained significant factors. Chemotherapy showed no association with ON. Due to the small number of events, the tests of significance as well as a multivariate analysis with more than two covariates can be considered only as tentative. A close correlation between teeth extraction after rt and the clinical beginning of an ON has been observed only in one patient. Two other patients with ON had teeth extractions after rt, but the radiographs showed definite evidence of an ON already before the extractions. For many years, teeth extraction in heavily irradiated mandibula is strongly discouraged and performed only if absolutely necessary.

The highest cumulative risk of grade 3 or higher ON has been observed in patients treated with a target dose per fraction of 2.00 Gy and a maximal mandibular dose per fraction of 2.20 Gy. Total target dose in this group was between 66.0 and 70.0 Gy (mean 69.30 Gy) with a maximal total dose to the mandibula between 66.0 and 79.7 Gy (mean 74.5 Gy): actuarial incidence of a grade 3 or higher ON in this group was 18% after 5 years (25% after 8 years) and was significantly higher compared to patients treated with a target dose of 2 × 1.20 Gy per day and total target doses between 72.0 and 76.8 Gy, with a maximal total dose to the mandibula between 74.4 and 82.30 Gy (mean 77.8 Gy): 5-year actuarial incidence of grade 3 or higher ON in this group was 2.4%.

## 4. Discussion

### *4.1. Incidence of ON*

Comparable to the published figures, ON were observed only after irradiation of the mandibula with a total dose exceeding 65 Gy applied with standard fractionation. The latency period between rt and the clinical dignosis of an ON in our patients was somewhat longer compared to the published figures: according to Murray et al. [7,8], 61% of all ON were diagnosed in the first year and 94% were apparent in the first 3 years after rt compared to 40 and 70% in our patients.

Considering all grades and only patients with mandibular resection, the cumulative risk on ON in our patients appears somewhat higher than in several published reports. But most of these reports include only the crude incidence rate with the total number of treated patients as 100%: in an analysis of patients with rt for cancer of the oral cavity or the oro- or nasopharynx from the MDA Hospital, Bedwinek et al. [1] observed a total incidence of ON of 1.8% after a total target dose of between 60 and 70 Gy, and an incidence of 9% after more than 70 Gy with approximately 10 Gy per week. In a more recent analysis of patients with tonsillar cancer from the MDA Hospital, Wong et al. [12] reported a total incidence of ON with mandibular resections of approximately 2% after a total dose of less than 67.5 Gy, and of 7% after higher doses. The overall 5-year survival in these patients was 70%. In a separate analysis of a selected group of patients from the MDA Hospital, who survived at least 5 years without recurrence, Larson et al. [5] reported an incidence of severe ON of 56.3%. It is not clear, however, what percentage of these patients from the treatment period 1964–1974 has been irradiated with a dose per fraction higher than 2 Gy, as indicated in some published data and dose response data from the MDA Hospital [3,10]. Perez at al. [9] reported a total incidence of ON with mandibular resection of 5% in 98 patients with an overall 5-year survival of 35%. Foote et al. [4] reported 2% mild and 2% severe ON in 50 patients after rt for a cancer of the base of the tongue.

### 4.2. Risk factors

According to our data, the risk of grade 3 or higher ON may vary significantly after doses per fraction between 2 and 1.2 Gy. Uni- and multivariate analysis with adjustment for the different lengths of follow-up showed a significantly lower risk after hyperfractionated rt with 2 × 1.2 Gy per day, and a total target dose between 72 and 79 Gy in approximately 6.5 weeks, compared to 1 × 2 Gy per day with a total target dose between 66 and 70 Gy in approximately 7 weeks. But, due to the small number of events, we cannot exclude other associations as explanation for the higher incidence of ON after 2 Gy per fraction (e.g., more frequently teeth in exposed mandibula). As noted above, there was no change in the pre-irradiation dental treatment policy, which has been performed in the dental department of our hospital. There was also no change in the use of hyperbaric oxygen between the different fractionation groups. There was only one patient (in the 2.0-Gy per fraction group), who had an inadequate dental extraction before the onset of ON.

In addition to the dose per fraction and the total dose, the state of the dentition, the association of the tumour to the bone, and the volume of the exposed mandibula are discussed as further risk factors.

In our patients, ON with mandibular resections have been observed only after inclusion of some part of the horizontal ramus, and were more frequent if more than half of the horizontal ramus was irradiated. Furthermore, the results of the surgical interventions were less satisfactory in patients in whom a larger part of the mandibula was included in the high dose volume.

A higher risk of ON after rt of tumours with a close association to the bone has been observed also from Murray et al. [7,8]: incidence of ON was 56.3% for tumours related to the bone compared to 28.8% for tumours adjacent to bone and 13.4% for other tumours. Despite the influence of a close association between the tumour and the bone, we did not observe an influence of the T stage on the risk of ON in our data. According to the published data and according to the higher risk of ON after irradiation of tumours with a close association with the mandibula, we would expect a higher risk in stage T4 due to bone infiltration. But there were only three patients with a stage T4 and bone infiltration in our study group.

According to the data of Morrish et al. [6] and of Murray et al. [7,8], 11.9 and 12.2% of the edentulous patients developed an ON, compared to 24% in dentate patients. Edentulous patients with extractions shortly before rt had a risk of ON comparable to the other edentulous patients. It is important, to eliminate sharp bone edges or bony irregularities and projections on the mandibular bearing surface as well as to wait after extractions for the healing of the mucosa covering. Extractions after high dose rt have a high risk of ON (61.1%) in the data of Morrish et al. [6].

## 5. Conclusions

The 8-year cumulative risk of ON of the mandibula with the necessity of resection after external rt with standard fractionation, and a maximal total dose between 70 and 77 Gy to a part of the horizontal ramus was approximately 25%. After hyperfractionated rt with 2 × 1.2 Gy per day, a 6-h interfraction interval and a total dose between 72 and 80 Gy, the risk of a grade 3 or higher ON was below 5%. Further risk factors are: a close association of the tumour with the bone and the status of the dentition. Before rt, specialized dental treatment is important.

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